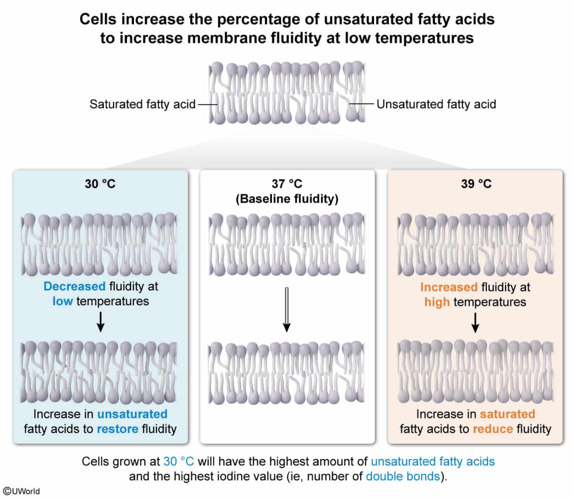
A scientist cultured samples of a cell line at three different temperatures: 30 °C, 37 °C, and 39 °C. She then extracted and isolated the membrane phospholipids of each sample and measured the iodine value, which quantifies the number of pi bonds that react with halogens such as iodine. Which sample most likely has the highest iodine value?

- A. The sample from cells grown at 30 °C [38%]
- B. The sample from cells grown at 37 °C [9%]
- C. The sample from cells grown at 39 °C [36%]
- D. All samples should have the same iodine value. [16%]

Correct
37% answered correctly

Explanation:



The fluidity of cell membranes is affected by several factors, such as temperature and membrane composition. **Increasing the temperature** increases the average kinetic energy of the membrane components, and therefore **increases fluidity**. Conversely, **decreasing the temperature decreases membrane fluidity**.

Under prolonged exposure to an external stimulus that affects membrane fluidity (eg, maintaining cell cultures at different temperatures), cells may initiate a homeostatic response **to counter this change**. For example, cells can respond to changes in membrane fluidity by changing their **membrane composition**.

The fatty acyl tails of membrane phospholipids can be either **saturated** or **unsaturated**. Saturated fatty acids have no double bonds (ie, they are saturated with hydrogens) and adopt an extended anti conformation that allows them to **pack tightly** with other saturated fatty acids. This tight packing provides a large surface area for intermolecular London forces, and therefore membranes enriched in saturated fatty acids exhibit **decreased fluidity**. Conversely, **unsaturated fatty acids** include at least one double bond, **which contains a pi bond**. Most double bonds in fatty acid tails are *cis* double bonds, which introduce a **kink** in the fatty acid tail. This kink **prevents tight packing** and reduces the strength of the intermolecular forces, and therefore **increases membrane fluidity**.

The question asks which sample has the **highest iodine value**. The iodine value reports the amount of pi bonds and correlates with the sample that has the **highest amount of unsaturated fatty acids**. Because unsaturated fatty acids increase fluidity, they would be upregulated as a **homeostatic response** to decreased fluidity at lower temperatures. Therefore, the **sample from cells grown at 30 °C will most likely have the highest iodine value**.

(Choices B and C) These samples were grown at temperatures higher than 30 °C, and therefore have higher temperature-induced fluidity. Consequently, less need exists for the homeostatic upregulation of unsaturated fatty acids.

(Choice D) Because the cells were cultured at different temperatures, they would have had sufficient opportunity to alter membrane composition as a homeostatic response to temperature-induced membrane fluidity changes. Therefore, the iodine value is likely to have changed for each sample.

Educational objective:
Unsaturated fatty acids are components of cell membranes that contain pi bonds. Unsaturated fatty acids increase membrane fluidity and are upregulated in homeostatic response to external temperatures that decrease fluidity.

Time spent: 0 sec	**QID:** 404061
Subject: Biochemistry	**System:** Carbohydrates, Nucleotides, and Lipids